

```

timescale 1ns / 1ps
`default_nettype none
`include "parameters.vh"
`include "maindec.v"
`include "alu_dec.v"

module control_unit(
    input wire clk,
    input wire rst_n,
    input wire [31:0] i_Instr,
    output wire o_RegSrcE,
    output wire o_Sel1E,
    output wire o_Sel2E,
    output wire [3:0] o_alu_ctrl,
    output wire o_LoadE,
    output wire o_BranchE,
    output wire o_JalE,
    output wire o_JalrE,
    output wire o_MemSrcE,
    output wire o_AMOEnE, // Atomic Memory Operation enable
    output wire [1:0] o_ResultSrcE
);

maindec md(
    .clk(clk),
    .rst_n(rst_n),
    .i_opcode(i_Instr[6:0]),
    .o_RegSrcE(o_RegSrcE),
    .o_Sel1E(o_Sel1E),
    .o_Sel2E(o_Sel2E),
    .o_LoadE(o_LoadE),
    .o_BranchE(o_BranchE),
    .o_JalE(o_JalE),
    .o_JalrE(o_JalrE),
    .o_MemSrcE(o_MemSrcE),
    .o_AMOEnE(o_AMOEnE), // New atomic operation enable signal
    .o_ResultSrcE(o_ResultSrcE)
);

aludec ad(
    .i_instr(i_Instr),
    .o_alu_ctrl(o_alu_ctrl)
);

endmodule

```